

HACKATHON 2026 • EIRGRID + CSO OPEN DATA

GridSense

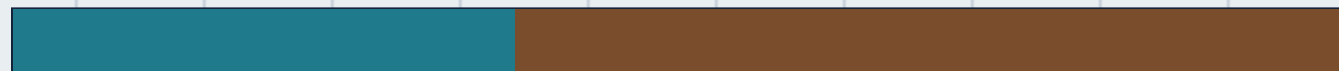
Forecast where Ireland's load will peak.
Route renewables there first. From Kerry to
Belfast.

THE PROBLEM

Ireland's wind is there. It doesn't reach the loads that need it.

37.6%

of island electricity was renewable, Oct 2025 – Oct 2026
(EirGrid, 15-min actuals, ROI + NI)



23.2%

of ROI metered electricity goes to data centres, as flat 24/7
baseload (CSO MEC02, 2025)

- On windy nights, wind farms are **dispatched down** because demand isn't there to absorb them.
- At the evening peak, gas fills the gap and the biggest loads run on it.
- Nobody can see **where** renewable power actually goes, by county, by hour.

The grid said no. The rules changed.

- **Jan 2022** EirGrid: no new data-centre connections in Dublin until 2028.
- **Apr 2024** AWS limits capacity in its Dublin region. GPU nodes "maxed out power-wise"; customers pointed to Sweden and other EU regions.
- **Aug 2024** Google's Grange Castle expansion refused over grid capacity (overturned on appeal, Sep 2026).
- **Dec 2025** CRU's new policy: new data centres must bring their own dispatchable generation or storage, and source **80%** of annual demand from new Irish renewables within 6 years.
- **Apr 2026** Amazon backs a €150m Midlands wind farm to power its data centres.

Where GridSense fits

- Operators must now **prove** how renewable their load is. GridSense measures it hour by hour.
- It forecasts where the next peak will hit and books recovered wind for it, easing the peak that triggered the moratorium.
- Councils and CRU get the same per-area numbers.

Sources: RTÉ / DCD, Jan 2022 • The Register via DCD, 10 Apr 2024 • Techerati, Sep 2026 • CRU decision via William Fry, 15 Dec 2025 • Irish Examiner, 27 Apr 2026

Forecast every region's peak. Route renewables to it first.

1 • Forecast regional peaks

Hourly load for each of 45 areas, 7 days ahead, from the trailing 8 weeks of real EirGrid data.

Backtest error **2.8%** ROI • **5.5%** NI



2 • Route renewables to them

Wind that would be dispatched down is scheduled, through network and storage capacity, into the biggest forecast peaks first.

Dublin City **696 MW** Tue 18:00
Cork County → Dublin City **485 MWh**



3 • Prove it, per area

Wind + solar share and CO₂ for any county or district, hour by hour, over 1, 3 or 12 months.

The moat: others report national averages after the fact. GridSense forecasts **where and when** each region will peak, at 2.8% error, and routes renewables there before it happens.

Electrons can't be steered. "Routing" means scheduling transmission, storage and renewable allocation against the forecast, so wind reaches the peak instead of being constrained off.

Same workload. More wind behind it.

- Your load stays exactly as it is. GridSense forecasts your peak and books renewables for it.
- The Dublin–Meath corridor's combined forecast peak: **1,845 MW** this week.
- Over 12 months the corridor gets **299 GWh** of recovered wind at its peak hours, cutting **33.8 kt** CO₂ from a 2.36 Mt footprint.
- The forecast retrains every day, so routing keeps up as your load grows.

What the customer gets

- Hourly-matched evidence for **24/7 carbon-free energy** claims
- Location-based **Scope 2** figures per site
- An answer to CRU's 80% renewable rule, with no change to SLAs

Evidence for where the next wind farm and cable should go.

- Per-county demand growth set against local renewable supply.
- The **Dublin–Meath data-centre corridor** consumes. The **western wind belt** (Kerry, Mayo, Donegal) generates.
- Hard numbers to lobby for more wind and solar, and for grid reinforcement where demand is rising.

Questions it answers

- Which counties' demand is outpacing renewables?
- How much wind is lost to dispatch-down, and where?
- Where will next week's peaks hit, and how much wind can reach them?

Pick the site with the surplus, not the congestion.

- See which regions are net renewable generators: the west and north-west.
- Compare wind + solar share and dispatch-down across candidate counties before committing to a site.
- A site near surplus wind starts out greener and puts less strain on the grid.

For investors

Renewable %, forecast peak load and routing headroom for every county and NI district, in one view.

"How much of my power was renewable this month?"

- Pick your area on the map and see its wind + solar share for the last month, quarter or year.
- See your area's CO₂ footprint as a plain number.
- Next: per-Eircode views from smart-meter data, with consent.

Why it matters

Seeing the number builds trust in the transition and makes flexible tariffs (EV charging, heat pumps) an easy choice.

Observability first. Then measurable cuts.

+621 GWh

dispatched-down wind routed to forecast peaks

71.2 kt

CO₂ avoided

37.6% →

39.1%

island wind + solar share

- Every area gets an hourly CO₂ footprint from EirGrid's real carbon intensity.
- Corporates report against measured hourly data instead of annual averages.

Modelled figures. Real EirGrid demand, wind, solar, CO₂ and wind forecast data (Oct 2025 – Oct 2026). Assumptions: the area split (CS0 MEC03 consumption, ROI; population, NI); dispatch-down = wind forecast minus actual when renewables ≥ 60% (9.2% of ROI wind, close to EirGrid's ~10%); routing recovers 50% of it (the network-constraint share) into the 6 forecast peak hours, weighted by forecast peak².

Open data in, one static dashboard out.

EirGrid API

15-min demand, wind, solar, CO₂ intensity, wind forecast. ROI + NI.

CSO + Census

MECO2 data-centre consumption, MECO3 county consumption, NISRA populations, OSi/OSNI boundaries.



Python model

Area load split, per-area peak forecast, routing of recovered wind from wind areas (by capacity) to the biggest peaks, CO₂ accounting.



Dashboard

Leaflet map of all 45 areas plus Chart.js. A static site with no server.

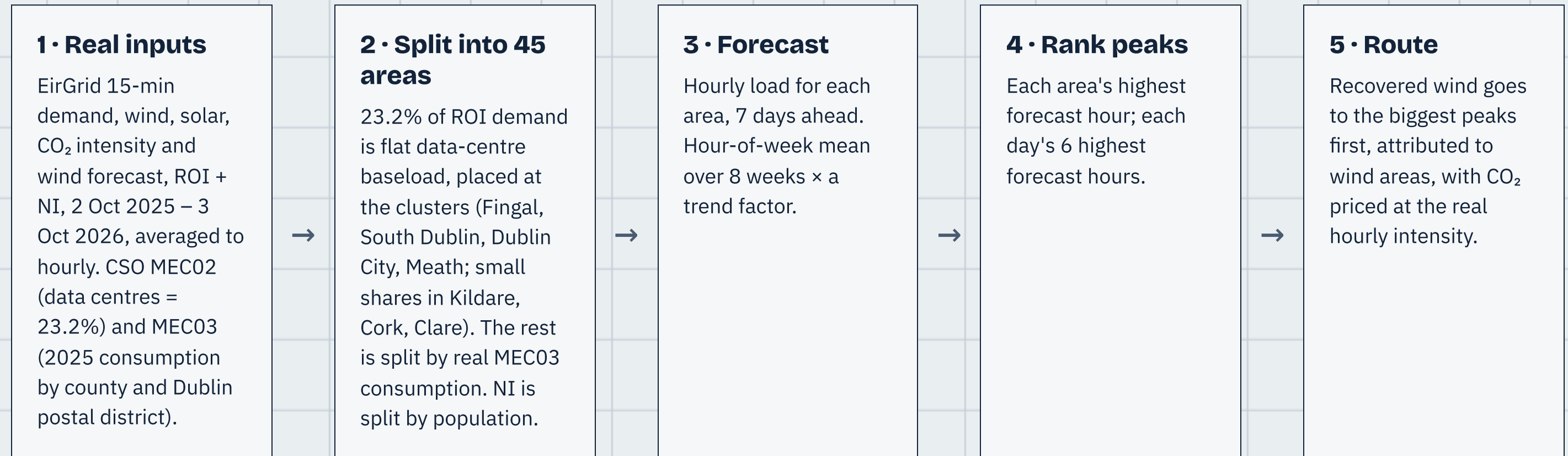
All public. All real.

SOURCE	WHAT WE USE
EirGrid Smart Grid Dashboard API	15-min demand, wind, solar, CO ₂ intensity, wind forecast. ROI + NI, 12 months
CSO PxStat MEC02	Data-centre share of metered electricity, quarterly, 2015–2025
CSO PxStat MEC03	Metered consumption by county and Dublin postal district, 2025
NISRA	Population by NI local government district
OSi / OSNI	County and LGD boundaries

Next data

- ESB Networks smart-meter data per Eircode, with consent
- EirGrid connected wind-farm list per node
- SEAI energy statistics

Five steps, Python standard library only.



Dublin postal districts are mapped to the four Dublin councils. City/county pairs (Cork, Galway, Limerick, Waterford, Tipperary) are split by population. MEC03's "Not coded" consumption (~30%, mostly large users) is left to the data-centre term. Check: the areas add up to EirGrid system demand within 1%.

A simple forecast, backtested on real weeks.

- 1. For each grid (ROI and NI), take the last **8 weeks** of hourly demand.
- 2. Average it by **hour of the week** (every Tuesday 18:00, and so on).
- 3. **Trend factor** = last week's actual ÷ what the averages predicted for last week.
- 4. Project **168 hours** ahead: hour-of-week average × trend.
- 5. Area forecast = its data-centre share + its MEC03 share of the regional forecast.

Why so simple: demand is strongly driven by time of week, and this method is transparent and hard to overfit. Upgrade path: weather inputs and gradient-boosted models.

2.8%

backtest error (MAPE), ROI

5.5%

backtest error (MAPE), NI: a smaller, noisier system

Backtest: forecast last week as if it hadn't happened yet, then compare with EirGrid actuals.

PEAK THIS WEEK	MW	WHEN
Dublin City	696	Tue 18:00
Fingal	503	Tue 18:00
South Dublin	395	Tue 18:00
Cork County	333	Tue 18:00
Meath	251	Tue 18:00

Book the wind for the biggest peaks first.

The pool

Estimated dispatch-down = EirGrid wind forecast – actual wind, counted only when the grid is $\geq 60\%$ renewable. That gives 9.2% of ROI wind and 12.2% of NI, close to EirGrid's reported $\sim 10\%$. We assume **50%** is recoverable by scheduling transmission and storage against the forecast (the network-constraint share).

Allocation

The pool goes into each day's 6 forecast peak hours, weighted by **forecast peak²**: biggest peaks get first call, but not winner-takes-all. Each area is capped at its fossil load in those hours, so it never goes past 100%. Leftover energy is re-shared among the areas still under their cap.

Sources and CO₂

Dispatch-down happens at wind farms, so each area supplies the pool in proportion to its wind capacity. This week: Cork County → Dublin City 485 MWh, Kerry → Dublin City 461 MWh, 12.8 GWh planned. CO₂ saved uses the **real carbon intensity of each peak hour** the wind displaces.

"Routing" means scheduling, not steering electrons. You can't send specific electrons to a specific address on a shared grid. GridSense plans transmission, storage and renewable allocation against the per-area forecast, then accounts for it hour by hour. Routing starts once 2 weeks of history exist.

How to read the dashboard.

VIEW	WHAT IT SHOWS
Header numbers	Island wind + solar share 37.6% → 39.1% with GridSense, 621 GWh routed, 71.2 kt CO ₂ avoided (12 months; they follow the period buttons)
1 / 3 / 12 months · Today vs GridSense	The period for every number; the same real demand and weather, without and with routing
Map: Forecast peak load	Default view. Darker navy = bigger peak in the next 7 days. This is the moat view
Map: Local generation ÷ use	Renewables generated ÷ used. Above 1× means the area exports wind (Kerry, Donegal, Leitrim): the sources
Map: Wind + solar share	Hour-matched share. Every area on one grid draws the same national mix (~32.9% over 3 months in ROI); GridSense makes the peak areas pull ahead
Routing lines	Wind areas → forecast peaks. Thicker = more energy. Hover a line for its MWh
Highest forecast load	Top 8 areas: peak MW, when it hits, wind routed to it. Click a row to select the area
Where the routed wind comes from	Source → peak flows in MWh, plus the biggest suppliers (Cork County 1.3 GWh, Kerry 1.3, Donegal 1.0)
View as	Data centre (1,845 MW corridor, −33.8 kt) · Government (80% target gap) · Siting new load · Household
Selected area	Share today → GridSense; orange value card, e.g. Fingal −9.3 kt CO ₂ (670.3 → 661.0 kt, 82.3 GWh more wind)
Charts	Weekly renewable / fossil / GridSense extra · 7-day forecast with backtest overlay · 48-hour routed wind landing in the peak hours

What we tried, what we changed, how we check it.

- **Forecast backtest:** 2.8% error ROI, 5.5% NI, on the last 7 days of real data.
- **Dispatch-down check:** our estimate gives 9.2% of ROI wind and 12.2% of NI, against EirGrid's ~10%.
- **Allocation:** greedy "biggest peak takes all" gave Fingal 0 GWh. We switched to peak²-weighted water-filling.
- **Area split:** population split gave Dublin City 596 MW and Fingal 565 MW. Real MEC03 consumption gives 696 and 503. Island totals were unchanged.
- **Pivot from flex to routing:** shifting data-centre load into windy hours gave 38.4%, 353 GWh, 48.6 kt. Forecast-led routing gives 39.1%, 621 GWh, 71.2 kt, and customers don't have to change anything.

Self-checks on every build

- Area demand adds up to system demand (within 1%)
- GridSense only adds renewables and cuts CO₂, never the reverse
- No area goes past 100% renewable
- The biggest peak gets at least as much as the 5th biggest
- The MEC03 split adds up to the CSO total
- Flows add up to what's routed

Data audit and UI tests

We investigated the identical-looking ~32.9% values, confirmed they're correct and relabelled the share as "wind + solar". A headless Chrome script clicks through every control after each change and checks for JS errors and the expected numbers.

What's modelled, and what judges ask.

Modelled, not measured

- How hourly load splits across areas (real CSO shares, but each area is assumed to follow the national hourly shape)
- County wind and solar capacity: approximate weights
- Dispatch-down: estimated from EirGrid forecast – actual
- The 50% recoverable share

v2 replaces each one: EirGrid dispatch-down reports and generator lists, ESB Networks smart meters per Eircode, live routing with EirGrid and SONI.

QUESTION	ANSWER
"You can't route electrons."	True. We route schedules: transmission, storage and renewable allocation planned against the forecast, then verified hour by hour.
"Why is every area ~32.9%?"	EirGrid only publishes generation for the whole grid. That's exactly the gap GridSense fills.
"Why such a simple forecast?"	2.8% error with a fully explainable baseline. Machine learning plus weather comes next.
"Is 621 GWh realistic?"	ROI wind is about 13 TWh a year, ~9% of it is dispatched down (~1.2 TWh), and we recover half: about 600 GWh.
"Who pays?"	Data centres, for 24/7 carbon-free and Scope 2 proof. EirGrid and the CRU, for connection and siting decisions. Later, green-hour tokens.

PROTOTYPE

Live demo

- Read the **Highest forecast load** ranking: where next week's peaks hit
- Click **Dublin City**, then **Fingal**, Ireland's data-centre hub
- Toggle **With GridSense routing** and switch to **12 months**
- Hover a routing line: **Cork County → Dublin City, 485 MWh** this week
- Read **Where the routed wind comes from**: Cork County 1.3 GWh, Kerry 1.3, Donegal 1.0; 12.8 GWh planned

From open data to live routing.

- **Now:** island-wide observability on real EirGrid data
- **Next:** per-Eircode load forecasts from ESB Networks smart meters
- **Then:** live day-ahead routing plans with EirGrid and SONI
- **Later:** hourly certificates for 24/7 CFE and Scope 2 reporting
- **Add-on:** flex signals for customers who want to go further

The ask

- Node-level wind data from EirGrid and SONI
- A data-centre pilot partner in Fingal or Meath
- A policy sponsor for forecast-led grid scheduling

Two bets beyond the prototype.

1 • Green-hour tokens

Rate-limit the island's forecast renewable output into hourly tokens, the same way an API rations requests.

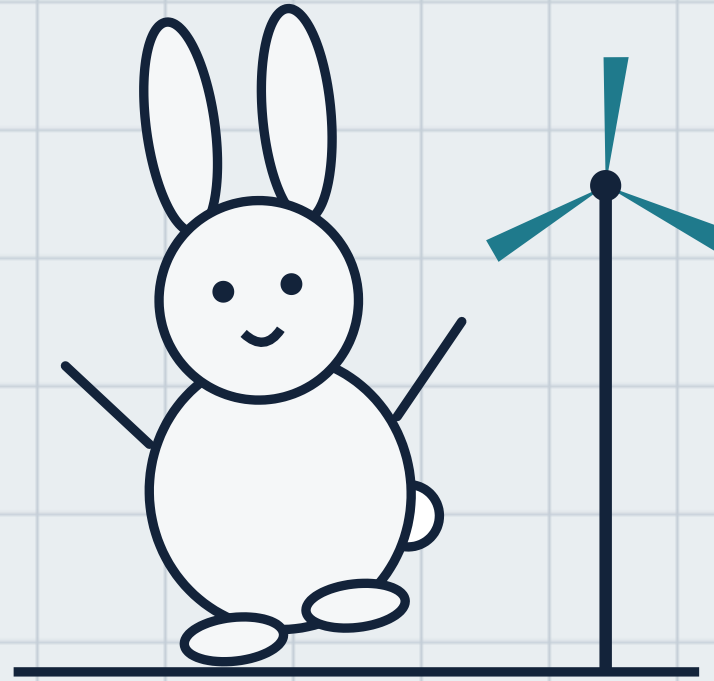
- Companies earn **priority access** to green hours by funding new renewables.
- A per-company cap stops any one buyer from cornering the wind.
- Unused tokens go back to the pool for everyone else.
- Every token spent is an auditable hourly record for CSR and Scope 2 reports.

2 • Micro-turbines on Ireland's canals

The Grand, Royal, Barrow and Shannon–Erne waterways and their feeders were built to manage water flow. That flow can turn small turbines too.

- Fit small turbines at locks, weirs and sluices, where the water drops.
- Unlike wind, the output is steady day and night.
- GridSense picks sites where local generation is lowest compared with demand.
- Partners: Waterways Ireland, local councils and community energy groups.

Canal water moves slowly, so most of the energy is at the drops (locks and weirs), not in the open channel. A pilot at one lock would show the real yield.



Thank you.

This bunny's moves are powered by wind
that would otherwise have been switched off.

Questions?